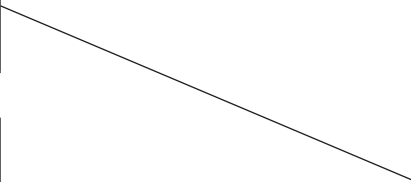


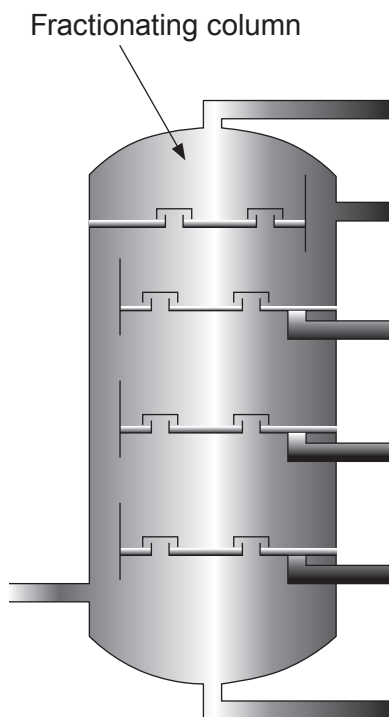
Answer **all** questions.

1. Raw materials are extracted from the Earth’s crust or atmosphere by different methods.
- (a) Identify the correct extraction method for each raw material by drawing a line.  
*One has been done for you.*

[2]

| Raw material            | Extraction method                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------|
| crude oil               |  fractional distillation |
| shale gas               | drilling                                                                                                   |
| haematite (a metal ore) | surface mining                                                                                             |
| oxygen                  | fracking                                                                                                   |

- (b) A fractionating column is used to separate the hydrocarbons in crude oil. The table shows information about some of the fractions obtained.



| Fraction       | Carbon chain length | Boiling range (°C) |
|----------------|---------------------|--------------------|
| Refinery gases | 1-4                 | <40                |
| Petrol         | 5-8                 | 40-110             |
| Naphtha        | 8-10                | 110-180            |
| Kerosene       | 10-16               | .....              |
| Diesel         | 16-20               | 260-320            |
| Fuel oil       | 20-50               | 320-400            |
| Bitumen        | >50                 | >400               |

(i) **Complete** the table. [1]

(ii) Name the process which is used to separate crude oil into its fractions. [1]

.....

(iii) State what happens to the crude oil immediately before it enters the column. [1]

.....

(iv) State the fraction that a hydrocarbon with a boiling point of 290 °C will belong to. [1]

.....

(c) Complete the statements below by underlining the correct term in the brackets.

(i) The temperature (**increases, stays the same, decreases**) going up the fractionating column. [1]

(ii) The boiling point of a hydrocarbon (**increases, decreases, stays the same**) as the chain length increases. [1]

(d) The fractionating column produces hydrocarbons that can be further processed and used to make plastics.

**Complete the table** to show how polyethene, polychloroethene (PVC) and polytetrafluoroethene (PTFE) are made. [3]

| Monomer           |                   |                                                                                                              | Polymer    |                                                                                                          |
|-------------------|-------------------|--------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------|
| Name              | Molecular formula | Structural formula                                                                                           | Name       | Structure                                                                                                |
| Ethene            | $C_2H_4$          | $  \begin{array}{c}  H & & H \\  & \diagdown & / \\  & C = C \\  & / & \diagdown \\  H & & H  \end{array}  $ | polyethene |                                                                                                          |
| Chloroethene      | $C_2H_3Cl$        |                                                                                                              | PVC        | $  \left[ \begin{array}{cc}  Cl & H \\    &   \\  -C & -C- \\    &   \\  H & H  \end{array} \right]_n  $ |
| Tetrafluoroethene |                   | $  \begin{array}{c}  F & & F \\  & \diagdown & / \\  & C = C \\  & / & \diagdown \\  F & & F  \end{array}  $ | PTFE       | $  \left[ \begin{array}{c}  F \\    \\  -C- \\    \\  F  \end{array} \right]_n  $                        |